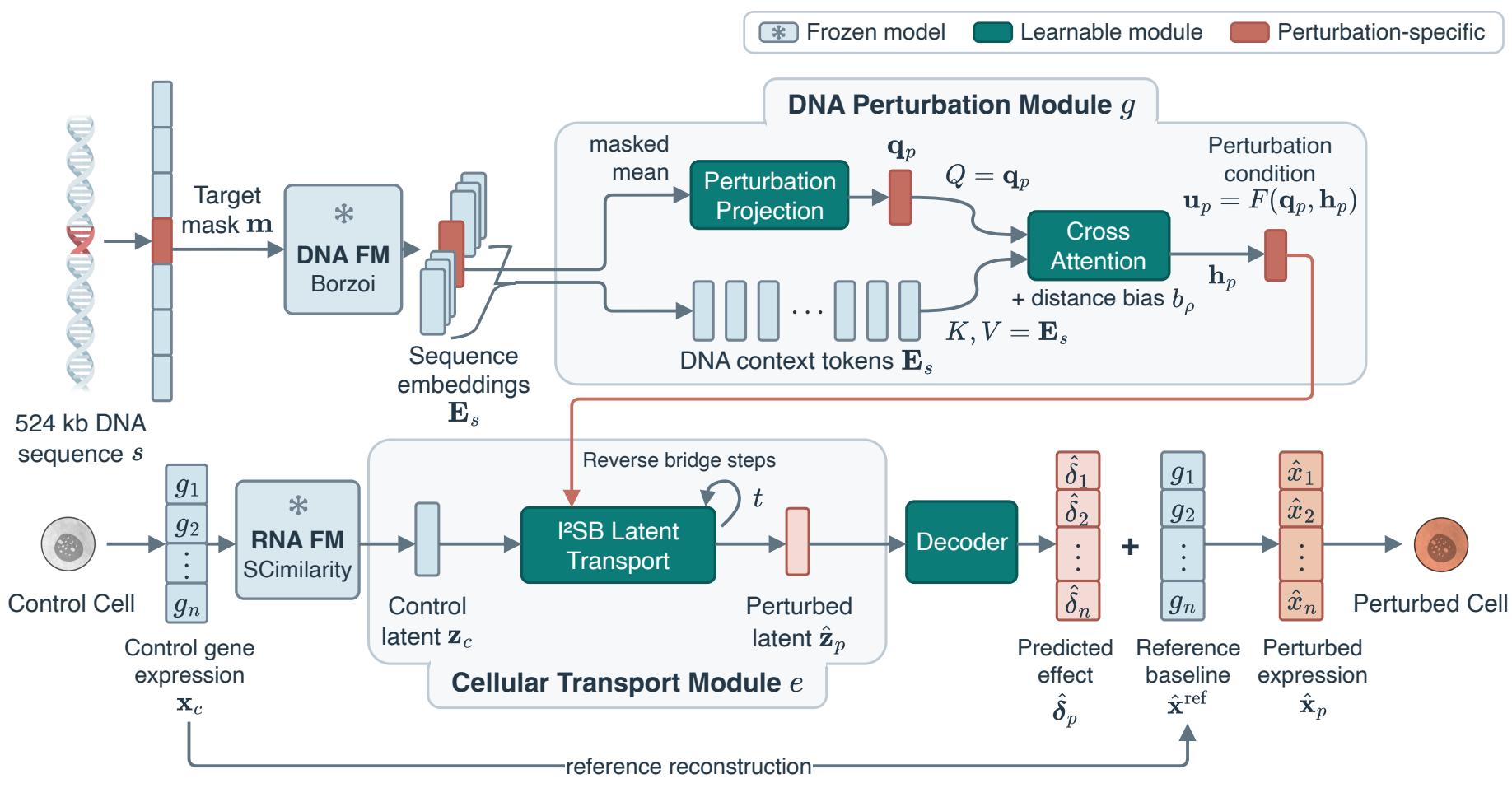

Frozen model
Learnable module
Perturbation-specific
DNA Perturbation Module g
524 kb DNA sequence s
Target mask $\mathbf{m}$
DNA FM Borzoi
Sequence embeddings $\mathbf{E}_s$
masked mean
Perturbation Projection
$\mathbf{q}_p$
$Q = \mathbf{q}_p$
Perturbation condition $\mathbf{u}_p = F(\mathbf{q}_p, \mathbf{h}_p)$
Cross Attention
$\mathbf{h}_p$
+ distance bias b_ρ
$K, V = \mathbf{E}_s$
DNA context tokens $\mathbf{E}_s$
Control Cell
g_1 g_2 g_n
RNA FM SCimilarity
Control gene expression $\mathbf{x}_c$
Reverse bridge steps
t
I²SB Latent Transport
Control latent $\mathbf{z}_c$
Perturbed latent $\hat{\mathbf{z}}_p$
Cellular Transport Module e
Decoder
$\hat{\delta}_1$ $\hat{\delta}_2$ $\hat{\delta}_n$
Predicted effect $\hat{\boldsymbol{\delta}}_p$
+
g_1 g_2 g_n
Reference baseline $\hat{\mathbf{x}}^{\text{ref}}$
$\hat{x}_1$ $\hat{x}_2$ $\hat{x}_n$
Perturbed expression $\hat{\mathbf{x}}_p$
Perturbed Cell
reference reconstruction